

Double-Plume Integral Models for Near-Field Mixing in Multiphase Plumes

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Abstract: We present a generalized integral model for multiphase plumes in stratified ambient conditions based on the double-plume approach, where the plume is composed of a rising, multiphase core plume surrounded by a counterflowing outer ring plume of dense fluid. The generalized model captures as limiting cases the current approaches in the literature, including two-fluid and mixed-fluid equations, continuous and discrete detrainment, dispersed-phase mass transfer, and two models for entrainment in the counterflow region. These modeling approaches are compared and validated against both laboratory and field-scale data. In unstratified conditions, all model formulations perform equally well. In stratification, entrainment in the counterflow region is best represented by correlation to the inner plume velocity instead of the difference between the inner and outer plume velocities. The vertical distribution of the exchange between the inner and outer plumes in the models differs from that measured in the prototype due to enhanced entrainment at the detrainment zone and forced entrainment from the collapsing intrusion layer. Nonetheless, the models predict well the length scales and volume fluxes at the detrainment zone and intrusion layer. Applications are demonstrated for reservoir air bubble plumes. The mass transfer and near-field mixing in the double-plume integral model prove sufficiently accurate to predict the depth of maximum plume rise (both the locations of total dissolution of the bubbles and the maximum height of the decelerating plume) and the volume flux, dissolved constituent mass flux, and trap height of the intrusion.

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Introduction

Integral models remain an important tool for design and analysis of multiphase plumes for environmental applications. Examples include installations of air bubble plumes for aeration, oxygenation, and destratification of lakes, reservoirs, and wastewater detention basins (e.g., Asaeda and Imberger 1993; Buscaglia et al. 2002; McGinnis et al. 2004; Sahoo and Luketina 2006; Singleton et al. 2007; Singleton and Little 2006; Wüest et al. 1992), analysis of the fate of natural methane gas released from deep sea hydrates (e.g., McGinnis et al. 2006), and of oil and natural gas released in accidental oil well blowouts (Yapa and Zheng 1997; Zheng et al. 2003), as well as design and environmental assessment of direct ocean disposal of carbon dioxide (e.g., Crounse et al. 2007; Liro

et al. 1992; Socolofsky and Adams 2005; Wannamaker and Adams 2006). These integral models of multiphase flows are based on the similarity assumption and entrainment hypothesis, which convert the three-dimensional problem to a set of one-dimensional equations. They are attractive because of their computational efficiency and because they have been successfully validated against diverse data sets. One difficulty for these models is the need to simulate the near-field recirculation induced by the plume, as in the case of a descending dense outer plume in stratified conditions (McGinnis et al. 2004). Solutions using a double-plume approach are available in the literature, where the recirculating flow is modeled as a descending annular ring plume (Asaeda and Imberger 1993; Crounse et al. 2007). An important uncertainty in the double-plume model is the entrainment between the counterflowing plumes, and work remains to evaluate these models for their reliability in estimating field-scale plume properties. In this paper, we develop a generalized double-plume integral model for multiphase flow that can be adapted to each of the models in the literature, and we evaluate both the model structure and model parameters by comparison to laboratory and field-scale data. This model comparison is insightful since each modeling scheme is based on different physics, and differences among their results help identify the physical mechanisms needed for an accurate model.

Integral plume models for multiphase flow in unstratified ambient conditions are simplified because the near-field recirculation does not play a dynamic role on the plume evolution. Original models are based on Kobus (1968) and Cederwall and Ditmars (1970), who incorporated the slip velocity between the dispersed and continuous phases into the single-phase models of Morton et al. (1956). Milgram (1983) verified the integral model approach

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for bubble plumes in unstratified conditions using laboratory and field-scale data. Because multiphase plumes are not strictly self-similar, the model parameters were shown to vary with height. Milgram found that the variation of the entrainment coefficient could be related to a local bubble Froude number. He also introduced a momentum amplification factor to account for mean momentum flux carried in the turbulence (Milgram 1983). Buscaglia et al. (2002) validated the integral plume equations by showing that they can be rigorously obtained for a dilute plume by laterally averaging the three-dimensional, mixed-fluid equations of motion. They compared the integral model results to computational fluid dynamic (CFD) simulations of the axisymmetric mixed-fluid equations. One limitation of the integral model was that it did not accurately predict the net buoyant force of the dispersed phase acting on the entrained fluid. Despite this small limitation, the integral model matched many aspects of the CFD plume characteristics, including the entrained fluid flux. Through these studies, integral models have been established as an effective means of analyzing multiphase plumes.

Because integral models of multiphase plumes in unstratified conditions predict well the mixing and momentum transfer in the plume, many bubble plume models for reservoir aeration and destratification follow an integral approach. An important component of such models is the mass transfer between the bubbles and the ambient fluid. Wüest et al. (1992) developed a discrete bubble submodel for a reservoir aeration plume, where the slip velocity, thermodynamics, and mass transfer rates are evaluated for a single bubble and then applied to all the bubbles in the flow at the same height. McGinnis and Little (2002) verified the discrete bubble model for unstratified bubble columns, and McGinnis et al. (2004) applied the submodel within a single plume integral model for aeration in a Swiss lake. The model results were compared to detailed vertical profiles of salinity, temperature, and oxygen. The data showed that the stratified plume fluid forms a downward-flowing outer plume that shrouds the inner plume, causing recirculation in the plume near field. The model comparison with the data showed that the integral model predicted the mixing and dissolution accurately and that the oxygen concentration in the plume matched the measured profiles if the near-field conditions were supplied as ambient data to the inner plume. Thus, while the dissolution and mixing in the inner core flow are well predicted by integral models, the near-field shortcircuiting caused by the stratification is not modeled and is needed to provide accurate concentrations in the plume.

An approach to capture this near-field recirculation is to model the outer flow as an annular ring plume of continuous phase fluid. McDougall (1978) introduced the concept of such a double plume based on observations of bubble plumes in linear stratification. Asaeda and Imberger (1993) showed that the upward-flowing components can be modeled as a single inner plume of mixed fluid and the downward flow as a single-phase outer ring plume. In their model, a discrete, fixed fraction of entrained fluid is removed from the inner plume to form the outer plume when the inner plume momentum flux approaches zero. Crouse (2000) and Socolofsky et al. (2002) introduced an algorithm for continuous detrainment based on a local force balance in the plume. Crouse et al. (2007) compared the continuous peeling model to laboratory data, and Wannamaker and Adams (2006) applied the model to negatively buoyant carbon dioxide plumes. In addition to the detrainment algorithm, a counterflowing entrainment model is needed to account for exchange between the inner and outer plumes. Entrainment in counterflow is not well understood, and the entrainment flux into the inner plume can be modeled as pro-

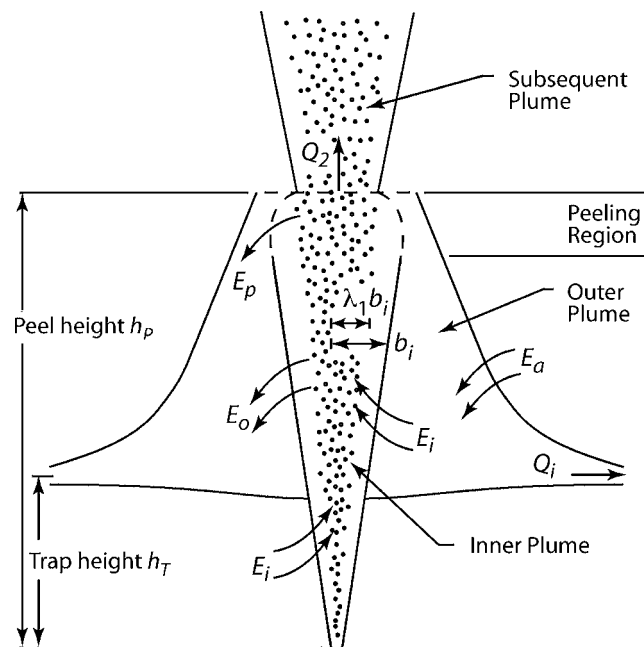


Fig. 1. Schematic of double-plume integral model for bubble plume in stratified ambient fluid. Several characteristic attributes of idealized bubble plume are identified including half width b_i and spreading ratio λ_1 of inner plume, intrusion layer flux Q_i , flux of entrained fluid escaping detrainment zone Q_2 , as well as important entrainment fluxes (E_i , E_o , E_a) and detrainment, or peeling, flux (E_p).

portional to either the velocity of the inner plume or the velocity difference between the inner and outer plumes. Each approach yields different recirculation, leading to a different plume near field and different behavior in subsequent plume structures forming above each detrainment zone. Thus, although the double-plume approach appears promising, the ability of the model to accurately predict the near-field mixing has not been completely investigated.

To address this shortcoming and to synthesize each of the modeling strategies in the literature, this paper presents a generalized integral model for multiphase plumes, paying special attention to the near-field recirculating flow. Several model formations, including multicomponent and mixed fluid approaches, discrete and continuous peeling, and two different entrainment models, are compared and validated against extensive data from the literature. The "Multiphase Plume Integral Model" section presents the general integral model equations. Each distinct model formulation is compared to laboratory and field data in unstratified and stratified conditions in the "Results and Discussion" section. Based on the results of the model validation, the "Application" section highlights some important insights for simulation, design, and operation of reservoir bubble plumes.

Multiphase Plume Integral Model

General Description

Fig. 1 shows a general schematic diagram of the integral model representation of a bubble plume in a stratified ambient fluid. Bubbles injected at the base of the plume rise, and their drag accelerates the surrounding fluid to form an upward-flowing, radially symmetric inner plume. To represent the fact that the

bubble column may be narrower than the velocity profile, a spreading ratio λ_1 is introduced to relate the plume half width b_i to the bubble column half width $\lambda_1 b_i$. Turbulence in the inner plume drives an entrainment flux E_i into the inner plume, first from the ambient fluid below the outer plume and later from the outer plume itself. Because the ambient fluid is stratified, the inner plume fluid eventually becomes negatively buoyant and begins to decelerate. When the inner plume fluid decelerates to zero upward velocity, it detains, or peels, from the plume at a height h_p and descends in the plume near field. This is represented in the integral model by a peeling region and formation of a descending, single-phase annular ring plume. The flux of fluid from the inner plume that escapes to form the outer plume is denoted by E_p , and a flux of fluid Q_2 remaining in the inner plume continues to rise upward with the bubbles. Because the bubbles escape the peeling region, subsequent inner plume structures form, and the double-plume structure repeats until the reservoir surface is reached or the dispersed phase fully dissolves.

As the outer plume descends, turbulence in the outer plume generates an entrainment flux E_a into the outer plume from the ambient fluid, and an entrainment flux E_o into the outer plume from the inner plume. The difference between E_i and E_o gives the net exchange flow between the counterflowing plumes. Because of the stratification, the outer plume also decelerates and eventually stops, at which point the integral plume model ends and an intrusion layer with fluid flux Q_i intrudes into the ambient fluid. In the unstratified case, the inner plume does not decelerate to zero, and no outer plume segments are required.

Strategies

The equations of fluid motion in an integral plume model are governed by the conservation of mass and momentum flux. The fluxes are obtained by integrating an assumed self-similar profile for the axial distribution of the state variables. Gaussian and top-hat profiles are commonly used; in the double-plume case, the top-hat profile is less ambiguous since the profiles do not extend to infinity as do Gaussian profiles. The conservation of momentum flux equation requires buoyant forces due to the bubbles and the entrained ambient fluid, which must be computed from appropriate closure schemes. Closure can be formulated through either a conservation of buoyancy flux equation (as in Asaeda and Imberger 1993), or directly through equations of state. Socolofsky et al. (2002) and Crounse et al. (2007) show that for multiphase plumes with complicated chemistry, such as carbon dioxide plumes, the conservation of buoyancy flux approach becomes cumbersome. The alternative approach is to track each component that affects density separately (i.e., concentration of dissolved components, salinity, and temperature) and compute the continuous and dispersed phase densities using an equation of state. Both approaches for the buoyancy flux yield the same evolution of the buoyancy.

For multiphase flow, the governing conservation equations can be formulated using either a mixed-fluid or two-fluid approach (Buscaglia et al. 2002). Most multiphase integral models in the literature follow the mixed-fluid approach, where a complete set of conservation equations is solved for a single-phase fluid whose properties represent the local average of the fluid mixture. The one exception is the set of continuous peeling models based on Crounse (2000), where a two-fluid type equation is used to represent the continuous phase momentum flux. Strictly speaking, the two-fluid approach solves a separate, complete set of conservation equations for all components in the flow. For multiphase

integral models, the mass and momentum conservation equations for the dispersed phase are typically replaced by single-bubble empirical relations for mass transfer and slip velocity. In addition, for dilute plumes, the conservation of continuous phase mass and volume fluxes can also be approximated without error using the mixed-phase approach. As a result, the models following Crounse (2000) are not strictly two-fluid models, but rather solve mixed-fluid equations where these approximations do not alter the two-fluid behavior of the model and introduce a two-fluid equation for momentum conservation where the mixed-fluid approach does alter the representation of the physics in the model. Thus, we will call these models hybrid two-fluid models.

Governing Equations

Our presentation of the governing equations follows closely the derivation in Crounse et al. (2007) but also includes generalizations to other models in the literature. Our generalized model is based on top-hat velocity profiles and makes the dilute-plume assumption, where $(1-\chi) \approx 1$; χ =local void fraction in the flow. We use the entrainment hypothesis to account for entrainment (Turner 1986) and, as in all other integral models, we assume self-similarity (see for example Milgram 1983). Though we develop both hybrid two-fluid and mixed-fluid equations for a dilute plume, only the conservation of momentum flux will have different terms depending on the approach.

For the conservation of mass flux, the inner and outer plume equations are

$$\frac{d(\pi u_i b_i^2)}{dz} = E_i - E_o - E_p \quad (1)$$

$$\frac{d(\pi u_o (b_o^2 - b_i^2))}{dz} = E_a + E_o - E_i + E_p \quad (2)$$

where u =continuous phase velocity; b =plume half width; and z =vertical coordinate, taken as positive upward; throughout this paper subscripts i , o , and a indicate the inner plume, outer plume and ambient fluid, respectively, and plume variables are taken to be functions of height. The terms on the right-hand side of the equations represent positive source fluxes per unit height (refer to Fig. 1). The entrainment fluxes are computed following Crounse (2000) as

$$\begin{aligned} E_i &= 2\pi b_i \alpha_i (u_i + c_1 u_o) \\ E_o &= -2\pi b_i \alpha_o u_o \\ E_a &= -2\pi b_o \alpha_a u_o \end{aligned} \quad (3)$$

where the α =entrainment coefficients and the signs provide that each flux is positive (note that the downward velocity u_o is negative). The coefficient c_1 allows different counterflow entrainment models to be simulated: for $c_1=0$, entrainment depends on the inner plume velocity only, and for $c_1=-1$, entrainment depends on the difference between the inner and outer plume velocities.

The peel flux E_p , which is due to buoyant detrainment from the inner plume, depends on whether a continuous or discrete peeling model is applied. For continuous peeling, E_p is given by the empirical relationship (Crounse et al. 2007; Socolofsky et al. 2002)

$$E_p = -c_2 \varepsilon \left(\frac{u_s}{u_i} \right)^2 \left(\frac{B_i}{u_i^2} \right) \quad (4)$$

where $c_2=1$; u_s =slip velocity of the dispersed phase; ε =calibration parameter; and B_i =local kinematic buoyancy flux of the continuous phase, defined as

$$B_i = g \frac{\rho_a - \rho_i}{\rho_r} \pi b_i^2 u_i \quad (5)$$

Here, ρ =density and we make the Boussinesq assumption with ρ_r as a constant reference density. In the discrete peel model (Asaeda and Imberger 1993), $c_2=0$ and E_p is replaced by a fixed fraction f of inner plume fluid which is detrained when the inner plume momentum flux approaches zero. Thus, the continuous peeling model allows peeling along the entire length of the inner plume, whereas the discrete peeling model only allows detrainment at the top of an inner plume segment.

The conservation of momentum flux for the inner plume depends on the whether the hybrid two-fluid or mixed fluid formulation is used. The generalized inner plume equation is

$$\frac{d(\pi u_i^2 b_i^2)}{dz} = \frac{\pi g b_i^2}{\gamma_i \rho_r} [c_3 (\lambda_1^2 \chi (\rho_a - \rho_b) + \lambda_2^2 (1 - \chi) (\rho_a - \rho_i)) + (1 - c_3) (\lambda^2 (\rho_a - \rho_m))] + E_i u_o - E_o u_i - E_p u_i \quad (6)$$

where $c_3=1$ gives the hybrid two-fluid model and $c_3=0$ gives the mixed-fluid model. Here γ =momentum amplification factor (Milgram 1983); λ_1 =ratio of the bubble column width to the velocity width (refer to Fig. 1), λ_2 =ratio of the concentration profile width to the velocity width; λ =ratio of the width of the mixed-fluid buoyancy flux profile to the velocity width; and the subscripts b and m denote the dispersed and mixed phases, respectively. The mixed-fluid density is computed as

$$\rho_m = \chi \rho_b + (1 - \chi) \rho_i \quad (7)$$

where the void fraction is given by

$$\chi = \frac{\dot{m}_b}{\rho_b \left(\pi \lambda_1^2 b_i^2 \left(u_s + \frac{2u_i}{1 + \lambda_1^2} \right) \right)} \quad (8)$$

$\dot{m}_b$ =local mass flux of the dispersed phase and u_s =slip velocity. The first three terms on the right-hand side of Eq. (6) represent the buoyant forces of the bubbles, the entrained fluid, and the mixture. By inspection, the hybrid two-fluid and mixed-fluid models are equal in unstratified conditions or when each of the λ s is taken as unity. The outer plume conservation of momentum flux is given by the single-phase equation

$$\frac{d(\pi u_o^2 (b_o^2 - b_i^2))}{dz} = \frac{\pi g (b_o^2 - b_i^2) (\rho_o - \rho_a)}{\gamma_o \rho_r} + E_i u_o - E_o u_i - E_p u_i \quad (9)$$

Note that the ambient fluid is assumed to be stagnant so that $u_a=0$.

In order that this model may easily account for buoyancy flux changes due to a wide range of physical, chemical, and thermodynamic interactions, we track the heat flux, salinity flux, and flux of dissolved constituents separately and compute the buoyancy using equations of state as in Crounse et al. (2007). The conservation of a conservative component of the flow for the inner and outer plumes is given by

$$\frac{d(\pi b_i^2 u_i C_{i,j})}{dz} = E_i C_{(a,o),j} - E_o C_{i,j} - E_p C_{i,j} - \frac{d\dot{m}_{b,j}}{dz} \quad (10)$$

$$\frac{d(\pi (b_o^2 - b_i^2) u_o C_{o,j})}{dz} = E_i C_{o,j} - E_o C_{i,j} - E_a C_{a,j} - E_p C_{i,j} \quad (11)$$

where the C =concentrations, and these equations must be solved separately for each component of interest j (e.g., heat, salinity, dissolved oxygen). In the inner plume equation, $C_{(a,o),j}$ =concentration of fluid entrained into the inner plume: $C_{a,j}$ when no outer plume is present and $C_{o,j}$ otherwise. The term $\dot{m}_{b,j}$ =time rate of change of the dispersed phase mass flux for constituent j and is computed by the discrete bubble submodel (see below).

Model Closure

To close the system of model equations given above, equations are needed to provide the mass transfer $\dot{m}_{b,j}$ and slip velocity u_s of the bubbles as well as the local density of the dispersed and continuous phases. To predict the mass transfer, we use the discrete bubble model introduced by Wüest et al. (1992). Our only deviation from the equations in Wüest et al. (1992) are for the calculation of the slip velocity, which we obtain following Crounse (2000) and Crounse et al. (2007). To compute densities, we use two equations of state: for gas bubbles, we use the air equation of state in Lemmon et al. (2000), and for water density, we use the seawater equation of state given in Gill (1982).

Initial Conditions

At the beginning of either an inner or outer plume segment, each of the state space variables must be estimated. For the properties of the fluid entrained at the source (i.e., density, salinity, concentration of dissolved species, etc.), the ambient properties can be safely assumed. For the velocity and plume half width, however, new equations valid in the zone of flow establishment must be supplied.

Diffuser

Wüest et al. (1992) introduced the densimetric bubble Froude number F as a means to estimate the initial velocity u_i of a multiphase plume at the level of the diffuser. F is given by

$$F = \frac{u_i}{\sqrt{2\lambda_1 R g (\rho_a - \rho_m) / \rho_m}} \quad (12)$$

where R =radius of the diffuser. Wüest et al. (1992) showed that $F=1.6$ provides realistic initial conditions at the source. Eq. (12) then defines a transcendental equation for the initial velocity since the void fraction needed to compute ρ_m depends on the velocity u_i . The initial plume half width is taken as the radius of the diffuser.

Subsequent Inner Plumes

For the continuous peeling model, the inner plume does not have any discontinuities, and no initial conditions are required for the subsequent plumes that form. When the discrete peeling model is used, the inner plume is terminated at the detrainment zone, and plumes forming above this region must be restarted. For these plumes, we propose using an F -type initial condition similar to Eq. (12), but we have the added difficulty of not knowing the initial radius. When we take 100% peeling, the subsequent inner

plume radius can be estimated as the width of the bubble column at the peel height. We estimate the growth of the bubble column using a turbulent diffusion model, and take

$$b_i = \sqrt{2Ez/u_s} \quad (13)$$

where z =height above the diffuser; E =turbulent eddy diffusion coefficient, specified using an analogy to channel flow so that $E = \eta Du_s$; η =constant coefficient; and D =characteristic length scale. Bombardelli (2004), Bombardelli et al. (2007) and García and García (2006) show that an appropriate parameter to scale bubble plume experiments between the laboratory and the field is given by

$$D = \frac{B}{4\pi\alpha^2 u_s^3} \quad (14)$$

which has units of length. We assume that the largest eddies in the bubble core will scale with D . We also select u_s as the velocity scale for the turbulent diffusion coefficient E since we have used this parameter throughout the derivation to represent bubble characteristics. By calibrating to the continuous peeling model, the factor $\eta = 1.5 \cdot 10^{-5}$ is obtained. This model has been tested from laboratory to reservoir scale plumes (water depth of order 0.1–100 m) and shows excellent scale independence.

When less than 100% peeling is used, the volume flux above the peel Q_i is known, and the plume half width can be computed from

$$b_i = \frac{Q_i}{\sqrt{\pi Q_i u_i}} \quad (15)$$

For either case, the initial velocity is computed using Eq. (12) with R replaced by the value of b_i obtained from either Eq. (13) or (15). Because F goes to zero at the peel, an initial value for the subsequent plume of $F=0.3$ is used to match the continuous peeling results.

Reservoir Surface

When a reservoir bubble plume reaches the free surface, all of the entrained fluid leaves the plume and forms a downward-flowing outer plume. At the surface, the detrainment process also engulfs additional fluid, taken in this model as a fixed fraction $c_4=0.1$ of the inner plume volume flux at the surface. An outer plume segment is then initiated with this initial volume flux. By analogy to Eq. (11), we define the outer plume Froude number as

$$F_o = \frac{u_o}{\sqrt{(b_o - b_i)g[\rho_a - \rho_o]/\rho_o}} \quad (16)$$

Using this form of the Froude number, an initial value of 0.1 provides a smooth variation of the state variables in the outer plume. Since the volume flux in the outer plume Q_o is known at the surface, the initial half width of the outer plume can be estimated from

$$b_o = \sqrt{\frac{Q_o^2}{\pi Q_o u_o + b_i^2}} \quad (17)$$

Thus, Eq. (16) is a transcendental equation for u_o .

Subsequent Outer Plumes

In the presence of stratification, the outer plume generated at the surface eventually stops near a level of neutral buoyancy. Outer plumes forming at the remaining detrainment zones must then be started independently. When the discrete peeling model is used,

the volume flux initializing the outer plume is known, and the initial conditions are computed just as in the section for the “Reservoir Surface” using Eqs. (16) and (17). When the continuous peeling model is used, however, the detrainment is spread out over a finite vertical region, and further work is required to identify where the outer plume will initialize. From laboratory experiments in stratification, the outer plume forms as a cascade of eddies that scale on the plume width (Socolofsky and Adams 2005). We therefore integrate the continuous peeling flux E_p over a vertical region equal to the local plume width to obtain an estimate for the initial volume flux Q_o in the outer plume. This volume flux is assumed to originate at the middle of the integrated region, and the initial velocity is computed using Eqs. (16) and (17). In regions where the peeling flux is very small (i.e., outside of the detrainment zone), this procedure results in an outer plume that immediately arrests to form an intrusion. To avoid this problem, we compute the right-hand side of Eq. (2) to determine whether the outer plume is strong enough to grow in size. If it is, a new outer plume segment is initiated; otherwise, the outer plume segment is discarded and a region of height b_i below this point is integrated and tested. This process continues down the plume until the diffuser is reached.

Numerical Solution

The governing equations presented in this section form a set of coupled, nonlinear ordinary differential equations (ODEs). Because of the discrete and sudden nature of the peeling process, these equations are also quite stiff—the solution requires either a very fine step size or an adaptive-step algorithm so that the peeling region is correctly identified and resolved. The results presented here were calculated using the second-order adaptive-step solver for stiff ODEs, **ode23s** in Matlab, based on the Runge–Kutta–Fehlberg method. Since both the inner and outer plumes are computed using an adaptive step size, linear interpolation is used to obtain results for both plumes at a given height.

Because the inner and outer plumes are calculated separately, iteration is required to obtain a converged solution. The plume usually reaches an equilibrium state after 3–5 iterations. We define convergence based on the number and locations of peels—iteration terminates when the relative error in each peel height is below 0.01.

Results and Discussion

Many of the models derived above (hybrid two-fluid, mixed-fluid, continuous peeling, discrete peeling, single-plume, and double-plume) have been compared to various data sets by previous authors. In this paper, we compare these models to both new data sets and each other. Table 1 summarizes the generalized multi-phase plume parameter values selected for each model we present. Values of c_1 , c_3 , λ_1 , and λ_2 were selected to give different model structures. The value of ε or f was then adjusted for each model until an optimal fit was obtained for the calibration data, as described in the following sections.

Unstratified Ambient Conditions

Fig. 2 compares the model results with laboratory and field-scale data from Milgram (1983) and Seol et al. (2007). In unstratified conditions, the length scale D given in Eq. (14) and the slip velocity u_s are the important parameters describing the flow. Bom-

Table 1. Parameter Values Defining Each of the Models Presented in the Model Comparison along with the Average Model Error for the Six Calibration Parameters in Fig. 3

Model number	Double-plume entrainment (c_1)	Two-fluid parameter (c_3)	Buoyancy spreading ratios		Detrainment model (ϵ)	f	Average RMS error ^a (%)
			λ_1	λ_2			
1	0.0	1	1.0	1.0	0.0150	—	33.0
2	-1.0	1	1.0	1.0	0.0175	—	36.4
3	0.0	1	0.8	1.1	0.0175	—	46.4
4	-1.0	1	0.8	1.1	0.0175	—	58.9
5	0.0	0	1.0	1.0	0.0150	—	33.0
6	0.0	1	1.0	1.0	—	<1 ^b	43.6
7	0.0	1	1.0	1.0	—	1.0	5,650

^aError reported as the root mean squared error between the simulation results and the regression curves reported in Socolofsky and Adams (2003) and (2005) normalized by the mean of the model results.

^bChosen as a parameter between 0 and 1 to match a desired level of detrainment.

bardelli et al. (2007) also show that the source conditions (diffuser size and design) affect the solution through a height $z \leq 5D$, above which the plume behavior becomes asymptotic. The model results were obtained for Model 3 (refer to Table 1) by varying the source flow rate for an air bubble plume with initial

bubble diameter of 0.01 m in a 100 m deep reservoir with mass transfer. For this size of bubbles, the gas mass flow rate reduces by about 60% over the height of the plume due to dissolution. The diffuser radius was 0.05 m, and two bubble flow rates were modeled: 2 and 50 NL/min. The effect of the initial conditions is also

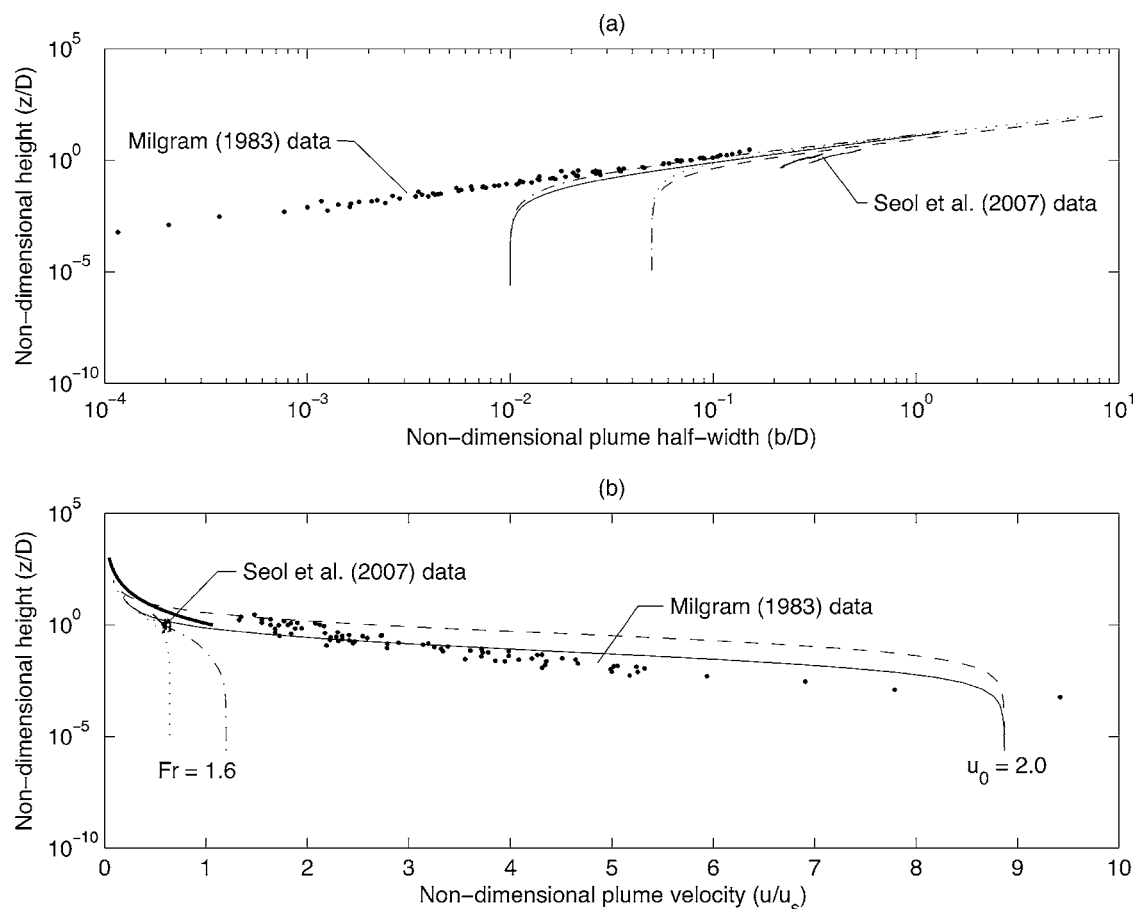


Fig. 2. Nondimensional comparison of top-hat integral model results to wide range of laboratory and field-scale data from Milgram (1983) and Seol et al. (2007); asymptotic solution from Bombardelli et al. (2007) is also shown as bold solid line in plot (b) for comparison. Model results are obtained for bubble plume with diffuser radius of 0.05 m in 100 m deep reservoir with mass transfer; initial bubble size was 10 mm in diameter. Results are shown for two gas flow rates and two different initial conditions. Thin solid line is for 50 NL/min gas flow rate and arbitrary initial fluid velocity of 2.0 m/s, dashed line is for 10 NL/min gas flow rate and arbitrary initial fluid velocity of 2.0 m/s, dash-dot line is for 50 NL/min gas flow rate and initial Froude number of 1.6 (fluid velocity of 0.27 m/s), and dotted line is for 10 NL/min gas flow rate and initial Froude number of 1.6 (fluid velocity of 0.15 m/s).

Table 2. Calibrated Values of the Entrainment Coefficients (α_i , α_o , and α_a) and Momentum Amplification Factors (γ_i and γ_o)

Parameter	Value
α_i	0.055
α_o	0.110
α_a	0.110
γ_i	1.10
γ_o	1.10

shown in the figure. The dotted and dash-dot lines show the results using an initial Froude number of 1.6, and the thin solid and dashed lines show the results for an arbitrary, higher initial velocity of 2.0 m/s.

The entrainment coefficient used in the simulations is that of Seol et al. (2007), who fit the relationship

$$\alpha_i = 0.18 \exp\left(-\frac{1.7u_s}{(B/z)^{1/3}}\right) + 0.04 \quad (18)$$

to their measured data and data from the literature, spanning the range $u_s/(B/z)^{1/3} = 0.3$ –25. The depth averaged value of α_i for plumes with maximum height of order D is close to 0.055, which is the value suggested by Asaeda and Imberger (1993) for a stratified bubble plume. Results using $\alpha = 0.055$ are quite similar to the results in the figure (within 2% for the velocity). Thus, constant-parameter models for the entrainment coefficient can also well predict the unstratified case.

Because there is no outer plume present in unstratified conditions, Model 4 (not shown) produces identical results; the other models having different λ_s (also not shown) give nearly identical results, but with slightly higher (+2%) velocities. Model 3 is closest to the measured data, which is consistent with best fit values of $\lambda_1 = 0.7$ –0.8 cited in the literature. The asymptotic solution for $z/D > 5$ in Bombardelli et al. (2007) is also shown as the bold solid line in the figure for comparison. Both the data and the integral model results converge on this asymptotic solution. Since the data and model results in the figure span a wide range of laboratory and field-scale data, these results show that the model captures the important scaling properties of the data.

Table 2 summarizes the model parameters for the inner plume found for the unstratified case using constant values of the model coefficients and for the outer plume using known values for single-phase plumes. The entrainment coefficients in Table 2 match those in Asaeda and Imberger (1993); the momentum amplification factors are similar to asymptotic values in Milgram (1983). We hold these values constant in the remainder of this paper in order to verify their applicability in stratified ambient conditions.

Stratified Ambient Conditions

The main model aspects requiring further evaluation in stratified ambient conditions are the continuous peeling model parameter ε and the type of recirculation entrainment (the value of c_1). Crounse et al. (2007) calibrate ε for the hybrid two-fluid model emphasizing agreement with the outer plume trap height h_T (refer to Fig. 1), the volume flux of detrained water entering the far-field intrusion Q_i , and the percent of fluid f peeled from the inner plume in the detainment zone based on data from Socolofsky and Adams (2003) and (2005). They also report limited new data on velocities in a counterflowing multiphase plume. Because of the scatter in the data, the counterflow velocity measurements could

not be used to select between entrainment models, and Crounse et al. (2007) report results for $c_1 = -1$ only. From their analysis, the calibrated value of ε was between 0.005 and 0.007.

Additional data in Socolofsky and Adams (2003) and (2005) help identify which model structures work well in stratification. Both papers report on experiments in a stratified tank 2.4 m deep. In addition to the parameters listed above, they also report values for the peel height h_p (refer to Fig. 1), the entrained fluid flux in the inner plume just below the first outer plume segment Q_1 , and the volume flux Q_2 (refer to Fig. 1) and negative buoyancy flux B_2 of entrained fluid escaping the first peeling zone. Socolofsky and Adams (2003) and (2005) also suggest an important plume scaling parameter given by the nondimensional slip velocity $U_N = u_s/(BN)^{1/4}$, where $N = \text{Brunt-Vaisälä buoyancy frequency } N = \sqrt{-g/\rho_r(\partial\rho/\partial z)}$ near the plume source.

To compare to the data in Socolofsky and Adams (2003) and (2005) and to simultaneously evaluate the model's ability to match the nondimensional plume scaling, we compute two different data sets and compare the results to the measured data. In the first data set, the model is run with the same inputs as were used in the physical experiments. These results are reported at U_N equal to 1.03, 1.44, 2.91, 3.67, and 4.68. For the second data set, the value of U_N is selected by using different slip velocities, buoyancy fluxes, and buoyancy frequencies than those measured in the laboratory experiments. The first group of artificial runs are at U_N equal to 0.94 and 0.93, where the slip velocity is adjusted to 0.05 m/s, $N = 0.30 \text{ s}^{-1}$, and the tank depth is preserved at 2.41 m. The other group of artificial runs are for U_N equal to 1.0, 2.0, 2.5, and 4.0, where the tank depth is taken as 5.0 m, $N = 0.35 \text{ s}^{-1}$, and the slip velocity is 0.15 m/s. In each of the artificial cases, the bubble flow rate is adjusted to give the desired values of U_N .

Runs for Models 1–4 (refer to Table 1), based on the continuous peeling and hybrid two-fluid approach, yield differing results, as presented in Fig. 3. Each model exhibits similar trends with increasing U_N . The trends are also similar to those in the measured data, with the greatest deviations for Figs. 3(a, e, and f). Model 1 consistently predicts the highest values; Models 2 and 3 predict very similar values for most parameters. Although the agreement varies by parameter, Model 1 fits the data the best. The other models underpredict the peel and trap heights and the fluxes escaping the first detainment zone. Only the intrusion layer flux is matched well by all models, Models 2 and 3 matching the closest. The optimum values of ε used to obtain these results are reported in Table 1 along with the average root mean squared error for each model across all six parameters shown in Fig. 3.

Fig. 4 shows the percent fluid peeled through the detainment zone for Models 1–4. The models match the measured data at $U_N = 1.0$ and 4.5, but underpredict f in the range $2 \leq U_N \leq 3.5$. The model values of ε were calibrated to best match the parameters in Fig. 3. The low calibrated values of f that we obtain are required in order to match the fluxes Q_2 and B_2 escaping the peeling region, and are lower than those selected by Crounse et al. (2007), who only calibrated to the intrusion layer flux Q_i . As a result, our calibrated value of ε is about a factor of two higher than that suggested by Crounse et al. (2007).

Since each of the Models 1–4 predict very similar detainment fluxes, differences among model results are due to their underlying structure. The primary model error is that the peel height is underpredicted [refer to Fig. 3(a)], which then feeds back on each of the parameters compared in Fig. 3 by reducing the height in the plume over which entrainment can occur. These model errors have two major sources. First, models based on the entrainment model $c_1 = -1$ have greater exchange from the outer plume into

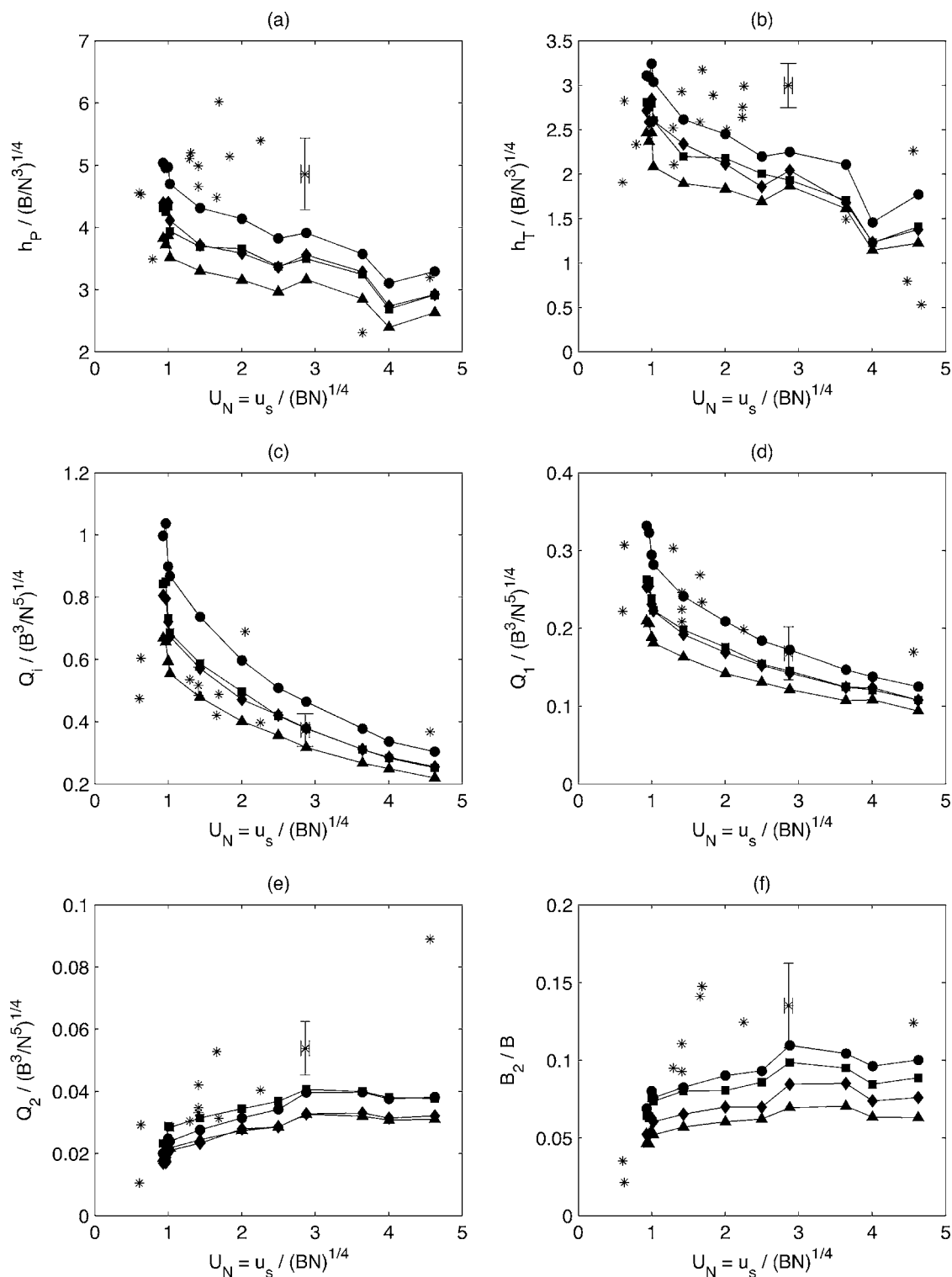


Fig. 3. Comparison of model simulations with data from Socolofsky and Adams (2003) and (2005). Stars are measured data with their reported error bars. Filled symbols are model results: Models 1, 2, 3, and 4, are given by circles, squares, diamonds, and triangles, respectively. Results are for first peel: h_p is peel height, h_T is trap height, Q_i is intrusion layer flux, Q_1 is entrained fluid flux below double-plume structure, Q_2 is entrained fluid flux escaping peel, and B_2 is absolute value of buoyancy flux of entrained fluid escaping peeling region.

the inner plume than models with $c_1=0$. Because this entrainment is from the counterflowing outer plume, the result is to reduce the velocity in the inner plume prematurely, explaining the lower peel height. Second, models with $\lambda_1 < 1$ underpredict the buoyant force of the bubbles. This is seen in Model 3 which also under-

predicts the peel height even though it is based on the weaker exchange model $c_1=0$. Crounse et al. (2007) argued that, especially in the counterflow region, it is unlikely that the entrained fluid width of the inner plume will be greater than the width of the bubble column; hence, we can expect $\lambda_1=1$ for a multiphase

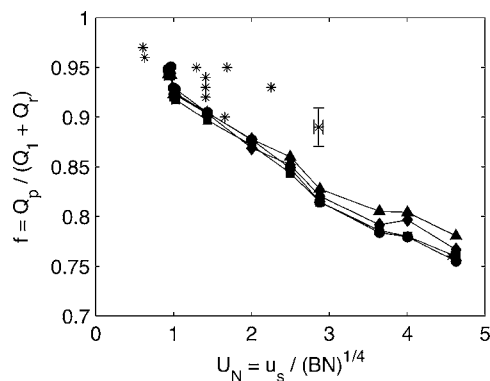


Fig. 4. Measured (stars) and modeled (filled symbols) results for fraction f of fluid peeled in first detrainment zone. Results are depicted for Models 1, 2, 3, and 4, using circles, squares, diamonds, and triangles, respectively. Data and error bars are from Socolofsky and Adams (2003).

plume in stratification. Thus, the better agreement for Model 1 is because it matches more accurately the exchange due to entrainment between the inner and outer plumes and better represents the buoyant force of the bubbles.

Fig. 5 shows selected results for the mixed-fluid Model 5 and the discrete peeling Models 6 and 7 compared to Model 1. The mixed-fluid Model 5 matches the results of Model 1 as expected, since the two models differ only if the λ s are different from unity. For the discrete peeling models, the results match those of Model 1 quite closely except for the prediction of Q_2 (not shown) and B_2 . For Model 6, which provides partial peeling ($f < 1$), the results for B_2 are close to Models 1 and 5. As expected for Model 7 with 100% peeling, both Q_2 and B_2 are predicted to be zero. By this comparison, Models 1 and 5 equally match the data, and the partial peeling Model 6 also appears acceptable.

Differences in the results between Models 1 and 6 are due to the selected value of f . The value of f for Model 6 is set as close as possible to the values calculated by Models 1–4 (refer to Fig. 4), though it remains a little high (3% higher). If the value of f is made smaller (generating less peeling), the negative buoyancy flux starting the next inner plume is too great, and the model self-compensates by immediately peeling a second time, effec-

tively resulting in 100% peeling at each detrainment zone. Thus, when the discrete peeling model is used with partial peeling, the model will naturally adjust itself if the peeling fraction is selected too low. This behavior limits the partial peeling Model 6 in its ability to match the measured data.

Despite the general good agreement of all these models with the data, there remain some differences between the actual physics of the near-field recirculation and the way in which the recirculation flow is modeled that account for the differences between the measured data and model results, especially in the range of U_N between 1.5 and 4. First, when the fluid in the real plume initially detrains, some fraction of ambient fluid is engulfed by the detraining eddies, yielding enhanced entrainment at the top of the outer plume. Second, when the outer plume finally comes to rest, it intrudes as a density current at a level of neutral buoyancy. This intrusion causes a forced entrainment of intruding fluid at the level h_T . Both of these processes are not included in the current models. The extra entrainment has not been included since the initial conditions for the outer plume are already not well known. The forced entrainment from the intrusion is not included because it violates self-similarity, required by the integral equations. Thus, the peeling and counterflow entrainment models must compensate for these omissions.

To illustrate the differences in the modeled and actual recirculation, Fig. 6 compares the model results with measured data through the first double-plume structure for experiment Sed2 reported in Socolofsky and Adams (2003) having $U_N = 1.7$. This figure compares the measured net upward volume flux (difference between the inner and outer plume fluxes) with results for Model 1. The measured data (dashed line) show a gradual reduction in the net volume flux throughout the region between $z = 0.3$ and 0.6 m above the source. This gradual reduction is due to the intrusion layer formation. The enhanced entrainment at the detrainment zone is also seen in the measured data as the steep positive slope just below the top of the peel height at $z = 0.75$ m. Model 1 predicts well the peel and trap heights, but overpredicts the intrusion layer flux (difference between the maximum and minimum net volume flux). Models 2 and 3 (not shown) match the intrusion layer flux, but underestimate the peel and trap heights. Model 4 (also not shown) nearly predicts two complete double-plume structures within this region. All integral models also predict a sudden intrusion formation as shown in Fig. 6 at $z = 0.35$ m

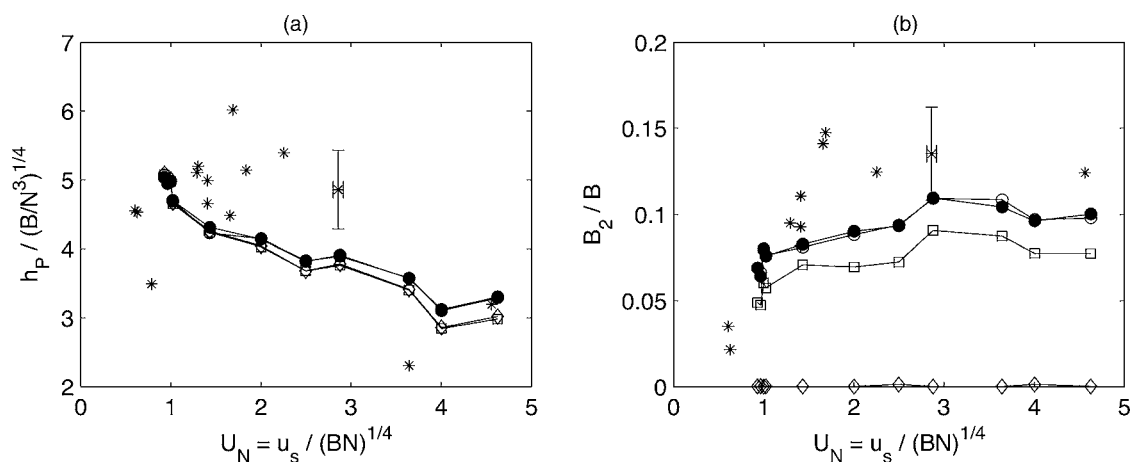


Fig. 5. Comparison of two-fluid model 1 (filled circles) with mixed fluid and discrete peeling Models 5, 6, and 7 (open circles, squares, and triangles, respectively). Stars are measured data with their reported error bars from Socolofsky and Adams (2003) and (2005). h_p is peel height, and B_2 is absolute value of buoyancy flux escaping first peeling region.

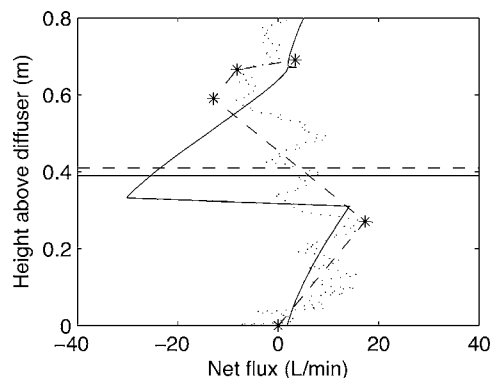


Fig. 6. Comparison of modeled net volume flux of entrained fluid with data measured by Socolofsky and Adams (2003). Measured data are for trap height (dashed horizontal line) and net volume flux (dotted line). Dashed line with stars shows smoothed net volume flux given in Socolofsky and Adams (2003). Model results are presented for Model 1 (solid line), showing net volume flux (variable with height) and modeled trap height (horizontal line).

height. In addition, none of the models show the enhanced de-trainment near the peel zone. Thus, while the double-plume model accurately captures the location of the peel zone (especially Model 1) and the total volume of fluid exchange between the inner and outer plumes (especially Models 2 and 3), the location of fluid exchange is not accurately represented. Since the elevation of fluid exchange is important in stratified fluids, this results in a somewhat incorrect estimate of the entrained fluid buoyancy which impacts the plume structure and makes it difficult for any one model to predict all plume characteristics equally well.

Although the locations of exchange between the inner and outer plumes are inaccurate in the model, comparison with field-scale data from McGinnis et al. (2004) for a reservoir aeration plume shows the utility of the double-plume integral model. McGinnis et al. (2004) present data for the net volume flux of an aeration plume in the Swiss Lake Hallwil in July 2001. Results of Models 1 and 6 are shown in Fig. 7. Comparing Figs. 6 and 7, the

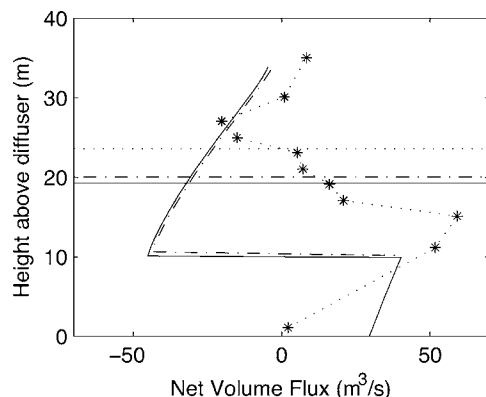


Fig. 7. Validation of integral plume model with field data for reservoir aeration plume from McGinnis et al. (2004). Measured net volume flux is given by stars and dotted line; measured trap height is estimated by current writers as location of zero net volume flux (dashed horizontal line). Model results are for Models 1 (solid line) and 6 (dash-dot line), showing net volume flux (variable with height) and modeled trap height (horizontal lines).

measured field data show very similar trends to the measured laboratory data. The total reservoir depth is 46 m above the diffuser; thus, the plume fully dissolves below the surface at $z = 35$ m. An important difference between the field and laboratory data is that the lake is closer to step stratification, with the thermocline at 33 m above the source. From Fig. 7, Models 1 and 6 both predict the maximum height of plume rise and the intrusion layer flux very well. The models agree with each other because of the lack of subsurface peels below the height of maximum plume rise. The trap height is somewhat underpredicted; however, the true trap height is unknown and was estimated by the current writers as the zero net-flux region in the measured data (compare to Fig. 6 for justification). The models also predict the vertical extent of the outer plume quite well, indicating dissolution is modeled correctly, and the models give a good estimate for Q_1 (indicated in the figure at $z = 10$ m). The remaining models show similar results, but underpredict the height of maximum plume rise and the height of the intrusion. Models 2 and 4 also underpredict the intrusion layer flux. Hence, Models 1 (or equivalently, Model 5) and 6 give the best comparison to the measured laboratory and field data, and reliably predict the length scales and bulk volume fluxes in field and laboratory plumes.

Application

Bubble plumes of air or pure oxygen are often used in reservoirs to overturn the reservoir in winter and to aerate the hypolimnion in summer. Asaeda and Imberger (1993) showed that destratification plumes are most efficient when the first peel occurs just below the reservoir surface. For a range of typical reservoir stratification levels using 20 mm diameter air bubbles, each of the models investigated here predicted the same air-flow rate needed to generate a peel just below the free surface. This is due to the step stratification which corrects the length scales in Models 2–4 and which arrests the outer plume in the epilimnion, thereby reducing the importance of the outer plume recirculation. Hence, for design of destratification plumes, all of the above models are acceptable.

For hypolimnetic aeration or oxygenation it is desirable not to significantly mix the stratification; thus, a much more accurate prediction of the plume interaction with the weakly stratified lake bottom water is required. Among the most important factors for design of aeration plumes are the rate of oxygen transfer, the maximum height of plume rise, the location of the trap height, and the net volume flux in the intrusion layer. Typical models in the literature apply the mixed-fluid approach without an outer plume structure. Fig. 8 shows the results using Model 1 with an outer plume and using Model 5 with the outer plume turned off, λ_1 set to 0.8, and ε set to zero in order to match the models of Wüest et al. (1992) and McGinnis et al. (2004). The density stratification was taken from the data for Lake Hallwil used in the previous section, and the ambient oxygen concentration was taken as constant at 0.001 kg/m^3 , or about 0.1 times the saturation concentration at the reservoir surface. From plots (a) and (c) in the figure, both models predict the same maximum height of plume rise and the same dissolution of the bubbles. The oxygen concentration at the top of the inner plume, however, is higher in the double-plume model because of recirculation in the outer plume. Plot (b) in the figure shows the oxygen concentration in the outer plume. The oxygen concentration at the base of the outer plume (0.0017 kg/m^3) is very close to the prediction at the top of the inner plume for the modified Model 5 (0.0018 kg/m^3). Thus, if

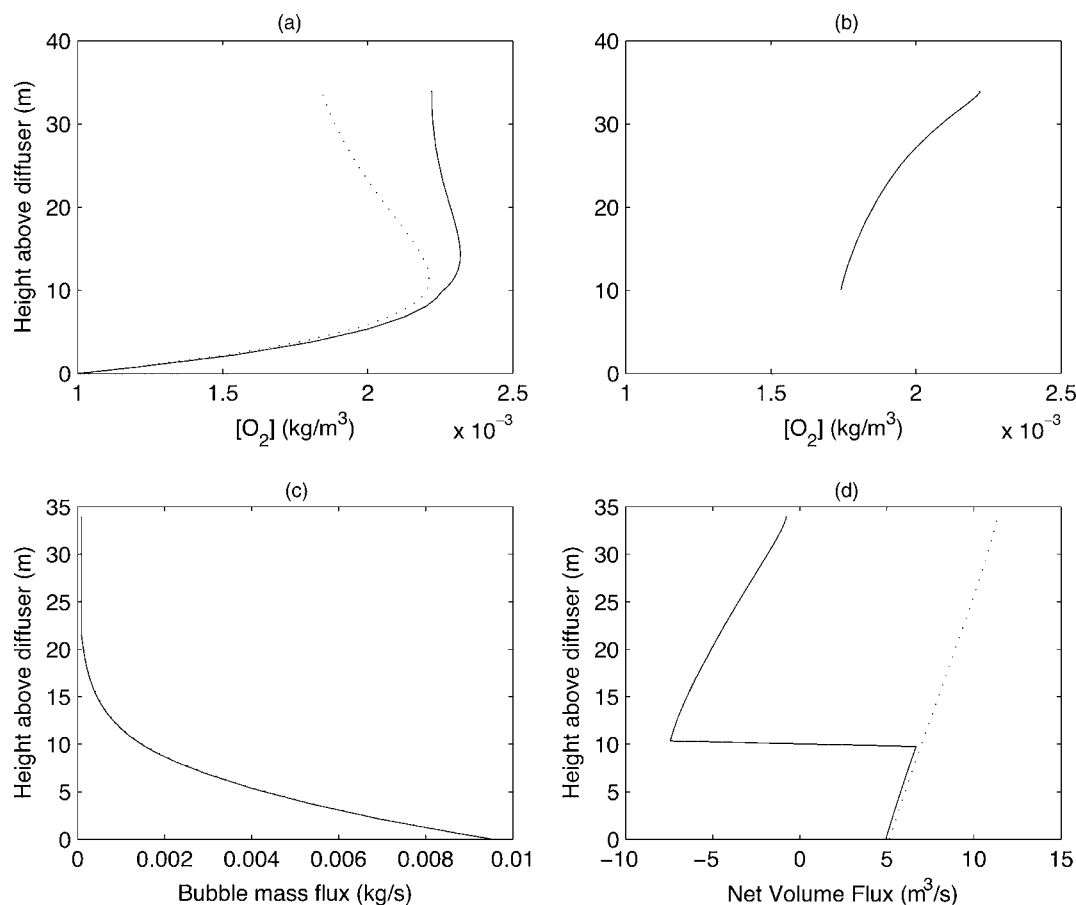


Fig. 8. Predictions for hypolimnetic oxygenation plume using Model 1 with double-plume structure (solid line) and Model 4 with outer plume turned off (dotted line). Plots (a) and (b) give dissolved oxygen concentration in inner and outer plumes, respectively. Plot (c) shows vertical variation of mass flux of bubbles; and plot (d) depicts net volume flux for each model.

we take these values as the oxygen concentration of water in the intrusion layer, both models predict the same concentrations. Plot (d) shows the prediction of the net volume flux in the plume. For the double-plume model, the intrusion layer flux is the difference between the maximum and minimum net flux in the plume, shown as $14.1 \text{ m}^3/\text{s}$. This value is also close to the volume flux at the top of the plume for modified Model 5 ($11.4 \text{ m}^3/\text{s}$). If we take the values at the bottom of the outer plume in the double plume model and the top of the inner plume in the single-plume model as predictions of the intrusion flux properties, both models give similar predictions to within 20% relative error. This close agreement is due to the fact that the single-plume model has been calibrated in the literature using these endpoints. If the ambient lake oxygen concentration is not constant, then differences in the rate of exchange between the plume and ambient fluid will cause greater differences in these models' predictions of the oxygen concentration in the plume. But because of the high partial pressure of oxygen in the air bubbles deep in the lake, these differences will not affect the rate of oxygen transfer. Hence, both model structures are acceptable for predicting the fate of air or oxygen bubbles injected in a lake, the double-plume model more accurately capturing the location and physical structure of the entrainment.

Summary and Conclusions

In this paper we compare several integral model formulations for multiphase plumes to laboratory and field-scale data and to each

other. The generalized equations for hybrid two-fluid and mixed-fluid models with continuous and discrete peeling are presented in the double-plume model approach, where detrainment fluid descends in a dense, outer ring plume. An important contribution to the governing equations is the specification of a consistent set of initial conditions for the subsequent inner plume structures and the outer plumes. This generalized integral model is then applied to investigate its ability to capture the near-field mixing induced by a multiphase plume.

From calibration to data in unstratified ambient conditions, several insights in the model structure were identified. Although multiphase plumes are not strictly self-similar and their parameters vary with height, most environmental applications have plumes with vertical extent on the order of $D=B/(4\pi\alpha^2u_s^3)$, such that constant parameter values can be used in modeling. The entrainment coefficient for the bubble core was found to be half the single-phase plume value, and the momentum amplification factor was chosen to have a constant value of 1.10.

Additional experiments in stratified ambient conditions provide a means to calibrate the remaining plume parameters. Models 1 (hybrid two-fluid approach) and 5 (mixed-fluid approach) performed the best overall. These two models are based on the $c_1=0$ entrainment model and have $\lambda_1=\lambda_2=1$. Both the continuous and discrete detrainment algorithms performed well. For the continuous peeling model $\varepsilon=0.015$, the discrete peeling algorithm has f set to the minimum value that generates a single peel at each detrainment zone. Both methods yield detrainment slightly less than that measured by Socolofsky and Adams (2003). This was

necessary to match the fluxes of entrained fluid and negative buoyancy measured to escape each detrainment zone. These fluxes in turn play a critical role in scaling subsequent plumes that form above each detrainment region.

The slight differences between the fluxes and plume length scales predicted by the models and those measured in the prototype result from errors in the physical representation of the exchange in the model. In real multiphase plumes, enhanced entrainment through the detrainment zone increases the initial outer plume flux, and forced entrainment from the intruding flow into the inner plume changes the vertical distribution of exchange between the inner and outer plume. Despite these differences, validation at the field scale shows that these models do well to predict the intrusion layer flux, the heights of the peeling zone, and the maximum plume rise, as well as the location of the intrusion.

Acknowledgments

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References

- Asaeda, T., and Imberger, J. (1993). "Structure of bubble plumes in linearly stratified environments." *J. Fluid Mech.*, 249, 35–57.
- Bombardelli, F. A. (2004). "Turbulence in multiphase models for aeration bubble plumes." Ph.D. thesis, Univ. of Illinois at Urbana-Champaign, Champaign, Ill.
- Bombardelli, F. A. B. G. C., Rehmann, C. R., Rincon, L. E., and Garcia, M. H. (2007). "Modeling an scaling of aeration bubble plumes: A two-phase flow analysis." *J. Hydraul. Res.*, 45(5), 617–630.
- Buscaglia, G. C., Bombardelli, F. A., and Garcia, M. H. (2002). "Numerical modeling of large-scale bubble plumes accounting for mass transfer effects." *Int. J. Multiphase Flow*, 28(11), 1763–1785.
- Cederwall, K., and Ditmars, J. D. (1970). "Analysis of air-bubble plumes." *KH-R-24*, W. M. Keck Laboratory of Hydraulics and Water Resources, Division of Engineering and Applied Science, California Institute of Technology, Pasadena, Calif.
- Crounse, B. C. (2000). "Modeling buoyant droplet plumes in a stratified environment." MS thesis, Massachusetts Institute of Technology, Cambridge, Mass.
- Crounse, B. C., Wannamaker, E. J., and Adams, E. E. (2007). "Integral model of a multiphase plume in quiescent stratification." *J. Hydraul. Eng.*, 133(1), 70–76.
- García, C. M., and García, M. H. (2006). "Characterization of flow turbulence in large-scale bubble-plume experiments." *Exp. Fluids*, 41(1), 91–101.
- Gill, A. (1982). *Ocean-atmosphere dynamics*, Academic, New York.
- Kobus, H. E. (1968). "Analysis of the flow induced by air-bubble systems." *Proc. 11th Int. Conf. Coastal Engineering*, ASCE, Reston, Va., 1016–1031.
- Lemmon, E. W., Jacobsen, R. T., Penoncello, S. G., and Friend, D. G. (2000). "Thermodynamic properties of air and mixtures of nitrogen, argon, and oxygen from 60 to 2000 K at pressures to 2000 MPa." *J. Phys. Chem. Ref. Data*, 29(3), 331–385.
- Liro, C. R., Adams, E. E., and Herzog, H. J. (1992). "Modeling the release of CO₂ in the deep ocean." *Energy Convers. Manage.*, 33(5–8), 667–674.
- McDougall, T. J. (1978). "Bubble plumes in stratified environments." *J. Fluid Mech.*, 85, 655–672.
- McGinnis, D. F., Greinert, J., Artemov, Y., Beaubien, S. E., and Wuest, A. (2006). "Fate of rising methane bubbles in stratified waters: How much methane reaches the atmosphere?" *J. Geophys. Res., [Oceans]*, 111(C9), C09007.
- McGinnis, D. F., and Little, J. C. (2002). "Predicting diffused-bubble oxygen transfer rate using the discrete-bubble model." *Water Res.*, 36(18), 4627–4635.
- McGinnis, D. F., Lorke, A., Wuest, A., Stoeckli, A., and Little, J. (2004). "Interaction between a bubble plume and the near field in a stratified lake." *Water Resour. Res.*, 40(10), W10206.
- Milgram, J. H. (1983). "Mean flow in round bubble plumes." *J. Fluid Mech.*, 133, 345–376.
- Morton, B. R., Taylor, G. I., and Turner, J. S. (1956). "Turbulent gravitational convection from maintained and instantaneous sources." *Proc. R. Soc. London, Ser. A*, 234, 1–23.
- Sahoo, G. B., and Luketina, D. (2006). "Response of a tropical reservoir to bubbler destratification." *J. Environ. Eng.*, 132(7), 736–746.
- Seol, D.-G., Bhaumik, T., Bergmann, C., and Socolofsky, S. A. (2007). "Particle image velocimetry measurements of the mean flow characteristics in a bubble plume." *J. Eng. Mech.*, 133(6), 665–676.
- Singleton, V. L., Gantzer, P., and Little, J. C. (2007). "Linear bubble plume model for hypolimnetic oxygenation: Full-scale validation and sensitivity analysis." *Water Resour. Res.*, 43(2), W02405.
- Singleton, V. L., and Little, J. C. (2006). "Designing hypolimnetic aeration and oxygenation systems—A review." *Environ. Sci. Technol.*, 40(24), 7512–7520.
- Socolofsky, S. A., and Adams, E. E. (2003). "Liquid volume fluxes in stratified multiphase plumes." *J. Hydraul. Eng.*, 129(11), 905–914.
- Socolofsky, S. A., and Adams, E. E. (2005). "Role of slip velocity in the behavior of stratified multiphase plumes." *J. Hydraul. Eng.*, 131(4), 273–282.
- Socolofsky, S. A., Crounse, B. C., and Adams, E. E. (2002). "Multi-phase plumes in uniform, stratified, and flowing environments." *Environmental fluid mechanics theories and applications*, H. Shen, A. Cheng, K.-H. Wang, M. H. Teng, and C. Liu, eds., ASCE, Reston, Va., 85–125.
- Turner, J. S. (1986). "Turbulent entrainment: The development of the entrainment assumption, and its application to geophysical flows." *J. Fluid Mech.*, 173, 431–471.
- Wannamaker, E. J., and Adams, E. E. (2006). "Modeling descending carbon dioxide injections in the ocean." *J. Hydraul. Res.*, 44(3), 324–337.
- Wuest, A., Brooks, N. H., and Imboden, D. M. (1992). "Bubble plume modeling for lake restoration." *Water Resour. Res.*, 28(12), 3235–3250.
- Yapa, P. D., and Zheng, L. (1997). "Simulation of oil spills from underwater accidents. 1: Model development." *J. Hydraul. Res.*, 35(5), 673–687.
- Zheng, L., Yapa, P. D., and Chen, F. H. (2003). "A model for simulating deepwater oil and gas blowouts—Part I: Theory and model formulation." *J. Hydraul. Res.*, 41(4), 339–351.